

Enterprise Project Management System (Zoho/OpenProject Inspired) - Implementation Plan

Phase 0: Architecture & Planning

Tech Stack:

Frontend: Next.js, TypeScript, Tailwind CSS, shadcn/ui, React Query.

Backend: NestJS, Prisma ORM, JWT Authentication, Swagger.

Database: PostgreSQL.

DevOps: Docker, GitHub.

Goal: Define architecture, coding standards, folder structure, and roadmap.

Phase 1: Foundation

Database Tables: users, roles, permissions, role_permissions, user_roles.

Features:

- Registration
- Login
- Logout
- Refresh Tokens
- Role Based Access Control
- Password Hashing (Argon2)
- JWT Security
- Input Validation
- Rate Limiting

Deliverables:

- Secure Authentication
- User and Role Management

Phase 2: Organization Module

Tables:

- organizations
- teams
- team_members

Features:

- Multi-organization support
- Team management
- Add/remove team members

Deliverables:

- Organization hierarchy
- Team management

Phase 3: Project Module

Tables:

- projects
- project_members

Features:

- Create projects
- Manage project members
- Project dashboard

Fields:

Name, Description, Start Date, End Date, Status, Owner

Phase 4: Task Management

Tables:

- tasks
- task_comments
- task_labels
- task_attachments

Features:

- Task CRUD
- Assignee and Reporter
- Priority and Status
- Comments
- Attachments

Statuses:

Todo, In Progress, Review, Done, Blocked

Phase 5: Kanban Board

Features:

- Drag and Drop task management
- Real-time updates

Library:

- dnd-kit

Deliverables:

- Interactive Kanban board

Phase 6: Milestones

Tables:

- milestones

Features:

- Milestone tracking
- Progress calculation
- Project completion tracking

Phase 7: Time Tracking

Tables:

- time_entries
- timesheets

Features:

- Start/Stop timer
- Manual time entry
- Weekly and monthly timesheets

Deliverables:

- Time logs
- Reporting

Phase 8: Issue/Bug Tracking

Tables:

- issues
- issue_comments

Features:

- Bug management
- Severity and Priority
- Assignment and resolution tracking

Phase 9: Notifications

Tables:

- notifications

Features:

- Real-time notifications
- Task assignment alerts
- Comment notifications
- Milestone notifications

Technology:

- Socket.IO

Phase 10: Reports & Analytics

Dashboard Widgets:

- Open Tasks
- Completed Tasks
- Overdue Tasks
- Hours Logged
- Project Progress

Library:

- Recharts

Phase 11: Audit Trail

Tables:

- audit_logs

Features:

- Track every important action
- Activity feed
- User accountability

Phase 12: Production Readiness

Security:

- CSRF Protection
- XSS Protection
- CORS
- Rate Limiting

Infrastructure:

- Docker Compose
- PostgreSQL Backups
- Logging

Testing:

- Unit Tests
- Integration Tests

Recommended MVP

Build these modules first:

1. Authentication
2. Organization Management
3. Projects
4. Tasks
5. Kanban Board
6. Time Tracking

These modules provide approximately 70% of the functionality found in Zoho Projects.

Folder Structure

project-management-system

frontend/
app/
components/
services/
hooks/
types/

backend/
src/

auth/
users/
organizations/
projects/
tasks/
timesheets/
issues/
notifications/
reports/
prisma/